Gabriel Vance
DSC630 – Predictive Analytics
Professor Andrew Hua
September 27, 2025

Individual Peer Evaluation Form

Gabriel Vance:

Write the name of your classmate you are preparing this review for in the designated column. Using a scale of 1-4 (1=strongly disagree; 2=disagree; 3=agree; 4=strongly agree) answer each question. If you aren't able to answer the question based on what is posted in the discussion board, reach out to your classmate for more information via the discussion board. Total the numbers in each column. **Make sure to answer the questions on the 2nd page.**

Evaluation Criteria	Peer Name: Nicholas Rodriguez
Has plan in place to complete course project.	4
Has found datasets/data sources to support project idea.	4
Has solidified project idea.	3
Has identified resources for project.	4
Topic is related to data science and demonstrates topics learned to date through program.	4
Risks and potential issues have been identified.	4
TOTALS	23

Feedback on Individual's project topic:

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1. How clear is the classmate's project topic? What questions does their topic make you consider?
    a. When I began reading the proposal, I felt like I knew what the goal was, but I had to go back and re-read the exact wording. I believe the goal is best summarized as "identifying potential risk factors with auto accidents". Nicholas goes into detail to describe the "how" for achieving this goal by mentioning the models he's going to use (Random Forest), testing parameters (cross fold, 80/20 split), and evaluation metrics (F1, recall, ROC-AUC). Nick has even found potential target variables such as total_claim_amount.
    b. Their topic makes me question the relevance of dataset features. I wonder if the features are sufficient or if more should be engineered to uncover new insights for analysis and prediction.

2. What risks or issues should your classmate consider while working on their project?

    a. They should be cautious of the training versus evaluation phase. Data must be completely isolated to prevent leakage.
    b. Nicholas has identified several models to use, such as K-Means and Random Forest Classifier. I eagerly anticipate observing how he constructs the models and how each model successfully completes a prescribed task.
    c. When conducting anomaly prediction, it is crucial to be aware that the classes will be severely imbalanced.

3. Additional suggestions/comments that might be beneficial to your peer?

    a. Clearly outline the problem using a bullet point and keep the goal front and center. It seems like the proposal split its focus between addressing regression and identifying anomalies.
    b. You've touched on classification, anomaly detection, regression, and k-means. I'd recommend using general tools for analyzing features and data and then picking one model to focus on for the main task. If the main goal is to

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predict the total_claim_amount, a linear regression model would be a great choice. I'm a bit worried about trying to achieve too many things at once by listing out so many models and goals.

c. I'd recommend starting with linear regression. Once you're happy with that, you can then tackle anomaly detection.

Adapted from a peer evaluation form developed at Johns Hopkins University (October, 2006)